

ASSESSMENT TEST: DATA ANALYST

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Business Problem

Management needs a simple view of trip value, payment behavior and high-value transactions.

Objective: identify operational opportunities that can be monitored with public trip data.

Scope: 200,000-row reproducible NYC yellow-taxi sample.

Data & Methodology

Source: NYC Taxi & Limousine Commission public trip records.

Fields: pickup/dropoff time, distance, locations, payment code and fare fields.

Processing: type correction → duplicate check → calculated duration/payment labels → quality flags
→ KPI aggregation.

Important: public-source data is an operational signal, not audited financial data.

Key Findings

Credit card is the dominant payment channel: 78.2% of trips.

Cash remains material: 14.8% of trips.

Average trip value is \$26.72; average distance is 3.582 miles.

Trips above \$50 represent 6.3% of the sample.

Deep-Dive Insight

Payment concentration makes payment reliability a high-leverage operational KPI.

High-fare trips are a smaller segment (12,675 trips) and can be monitored separately.

Management should avoid treating averages as sufficient: mix, outliers and data completeness need explicit monitoring.

Recommendations

1. Prioritize credit-card payment reliability monitoring.
2. Create an exception workflow for high-value / unusual fares.
3. Publish a recurring KPI pack covering trip volume, average trip value, distance and payment mix.

Expected Impact, Limits & Next Steps

Measure payment success/failure rate, exception resolution time and KPI variance.

Limitations: sample-based analysis, no customer-level context, no costs/profitability, and TLC cautions that submitted trip data may not be fully accurate.

Next step: rerun the provided Python pipeline on the full monthly TLC file and connect the outputs to Looker Studio.